ADCS TELEMETRY FEATURE ENGINEERING AND SIGNAL INTERPRETATION

Statistical Feature Reference Guide for Anomaly Analysis
ESA OPS-SAT FCNN Anomaly Detector ? RAG Knowledge Base
Version 1.0

1. INTRODUCTION

The OPS-SAT ADCS anomaly detection system uses 18 statistical and dynamic features extracted from raw telemetry segments to characterise the signal behaviour of each ADCS channel. This document explains what each feature measures physically, what values indicate nominal vs anomalous behaviour, and how to interpret feature combinations for diagnostic purposes.

Understanding these features is essential for explaining to operators WHY a segment was classified as anomalous by the FCNN model, and for grounding natural-language explanations in physical reality.

2. THE 18 FEATURES ? DEFINITIONS AND PHYSICAL MEANING

STATISTICAL MOMENT FEATURES:

FEATURE 1 ? MEAN (?)

Formula: ? = (1/T) ? x?
Physical meaning: The average value of the telemetry signal over the segment window. For magnetometers, this represents the average magnetic field component in that axis direction. For photodiodes, it represents average light intensity.

Anomaly indicators:
- Magnetometer: ? significantly above or below IGRF model prediction indicates sensor bias or orbital/attitude anomaly
- Photodiode: ? near zero when sunlit indicates sensor failure or pointing anomaly; ? near maximum in eclipse indicates ADC fault
- Normal range (magnetometer): ? varies between -60 and +60 ?T across orbital period; within any single segment, ? should be a smooth function of orbital position

FEATURE 2 ? VARIANCE (?²)

Formula: ?² = (1/T) ? (x? - ?)²
Physical meaning: The spread of signal values around the mean. High variance indicates rapid fluctuation (noise or rapid physical changes). Low variance indicates either a stable signal or a stuck/frozen sensor.

Anomaly indicators:
- Very HIGH variance ? noise, EMI, tumbling
- Near-ZERO variance ? stuck sensor, ADC fault, satellite in very steady attitude (rare for whole segment to be perfectly stable)
- Normal expected variance depends on window length and orbital phase

FEATURE 3 ? SKEWNESS (??)

Formula: ?? = (1/T) ? ((x? - ?)/?)³
Physical meaning: Asymmetry of the signal distribution. Positive skew means the signal has a longer tail toward high values; negative toward low values. Pure noise should be near zero skewness.

  Anomaly indicators:
  - Large positive or negative skewness in magnetometer ? asymmetric
    bias or clipping effect from saturation on one side
  - In photodiodes: large negative skew ? brief bright flashes (tumbling
    with Sun glints); large positive skew ? mostly dark with bright spikes

FEATURE 4 ? KURTOSIS (??)
  Formula: ?? = (1/T) ? ((x? - ?)/?)? - 3
  Physical meaning: "Tailedness" of the distribution. High kurtosis
  (leptokurtic) means heavy tails ? there are rare extreme values.
  Low kurtosis (platykurtic) means the distribution is very flat.

  Anomaly indicators:
  - High kurtosis in magnetometer ? impulse-like noise spikes (cosmic rays,
    SEU events, EMI transients)
  - Low kurtosis ? very uniform signal, possibly synthetic or frozen

RANGE FEATURES:

FEATURE 5 ? MINIMUM (x_min)
FEATURE 6 ? MAXIMUM (x_max)
  Physical meaning: The most extreme values reached by the signal.

  Anomaly indicators:
  - Magnetometer min or max outside ±65 ?T ? out-of-range anomaly
  - Photodiode max much lower than expected for sunlit condition ? degradation

FEATURE 7 ? RANGE (x_max - x_min)
  Physical meaning: Total excursion of the signal. Large range means the
  signal sweeps widely; small range means it stays confined.

  Anomaly indicators:
  - Near-zero range ? stuck value
  - Unusually large range ? large oscillation (tumbling, interference)
  - Range exactly equal to ADC full-scale ? saturation clipping

FEATURE 8 ? MEDIAN
  Physical meaning: The middle value when sorted. More robust to outliers
  than the mean.

  Anomaly indicators:
  - Large difference between mean and median ? asymmetric distribution
    suggesting one-sided bias or outlier contamination

DYNAMIC FEATURES:

FEATURE 9 ? ROOT MEAN SQUARE (RMS)
  Formula: RMS = ?[(1/T) ? x?²]
  Physical meaning: Energy content of the signal, including both mean and
  variation. Higher than expected means the signal is more energetic.

  Anomaly indicators:
  - Elevated RMS in magnetometer ? large bias or increased amplitude
    oscillation (reaction wheel vibration coupling, attitude oscillation)

FEATURE 10 ? MEAN ABSOLUTE DEVIATION (MAD)
  Formula: MAD = (1/T) ? |x? - ?|
  Physical meaning: Average deviation from the mean, robust to outliers.

Similar to standard deviation but less sensitive to extreme values.

Anomaly indicators:
- Elevated MAD ? sustained noise above normal floor

FEATURE 11 ? FIRST-DIFFERENCE MEAN (??)
Formula: ?? = (1/(T-1)) ? |x_{i+1} - x?|
Physical meaning: Average step size from one sample to the next.
This directly measures the rate of change of the signal.

Anomaly indicators:
- HIGH ?? ? rapid changes (noise, tumbling, fast oscillations)
- NEAR ZERO ?? ? stuck value or extremely stable signal (unusual)
- This feature is very powerful for distinguishing noise from bias:
  a biased signal has small ?? but large mean deviation from IGRF

FEATURE 12 ? LINEAR REGRESSION SLOPE
Physical meaning: Best-fit linear trend over the segment window.
A positive slope means the signal is trending upward; negative means
trending downward.

Anomaly indicators:
- Large positive or negative slope ? the signal is drifting systematically
  within this window. In magnetometers, moderate slopes are normal (the
  field changes with orbital position); unusually steep slopes indicate
  fast attitude change (manoeuvre, oscillation, or tumbling).
- In photodiodes, a large positive slope means going from shadow into
  light (eclipse exit); large negative slope means entering eclipse.

STRUCTURAL FEATURES:

FEATURE 13 ? ZERO-CROSSING RATE
Formula: ZCR = (1/(T-1)) ? ?[x? · x_{i+1} < 0]
Physical meaning: Fraction of consecutive pairs that cross through zero.
A high ZCR means the signal oscillates through zero frequently.

Anomaly indicators:
- Very HIGH ZCR ? rapid alternation (noise around zero, or high-rate
  tumbling causing magnetometer to swing through zero)
- Zero ZCR combined with stuck value ? the signal never crosses zero
  because it is frozen
- High ZCR in magnetometers during de-tumble is NORMAL

FEATURE 14 ? LAG-1 AUTOCORRELATION (??)
Formula: ?? = Corr(x?, x_{i+1})
Physical meaning: How similar each sample is to the previous one.
?? near 1.0 ? signal changes very slowly (smooth or stuck).
?? near 0 ? signal is random (white noise).
?? near -1 ? signal alternates between opposite values.

Anomaly indicators:
- ?? ? 1.0 AND low variance ? STUCK VALUE (sensor frozen)
- ?? ? 1.0 AND normal variance ? smooth physical signal (normal for
  magnetometers between OOL threshold and normal variation)
- ?? near 0 ? HIGH NOISE or rapid oscillation
- ?? < 0 ? alternating signal (rare in magnetometers; possible in
  noisy photodiodes during tumble)

FEATURE 15 ? SHANNON ENTROPY

  Formula: H = -? p(x) log? p(x) (over histogram bins)
  Physical meaning: Unpredictability or information content of the signal.
  High entropy ? signal is complex and hard to predict.
  Low entropy ? signal is simple, repetitive, or constant.

  Anomaly indicators:
  - Minimum entropy (?0) ? stuck value or highly regular periodic signal
  - Maximum entropy ? pure random noise
  - Normal magnetometer entropy is intermediate (smoothly varying field)

FEATURE 16 ? POWER SPECTRAL DENSITY PEAK FREQUENCY
  Physical meaning: The dominant oscillation frequency in the signal.
  For a smooth magnetometer signal this peaks at the orbital frequency
  (~1/5700 Hz for 95-min orbit). For a noisy signal it peaks at higher
  frequencies.

  Anomaly indicators:
  - PSD peak at high frequency ? noise or interference
  - PSD peak at a known frequency (e.g., reaction wheel rotation rate) ?
    mechanical vibration coupling
  - PSD peak at exactly 0 Hz ? constant bias with no oscillation

3. FEATURE COMBINATION PATTERNS AND THEIR DIAGNOSTIC MEANING
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PATTERN A: NOMINAL MAGNETOMETER SEGMENT
  ?: moderate (matches IGRF)
  ?²: low-moderate (smooth field variation)
  ??: HIGH (0.90+)
  ??: LOW
  ZCR: LOW
  H: moderate
  ? RESULT: Normal telemetry, no anomaly

PATTERN B: BIAS ANOMALY
  ?: ELEVATED or DEPRESSED significantly
  ?²: near-normal (bias doesn't add noise)
  ??: HIGH (still smooth signal, just offset)
  ??: similar to normal
  ? RESULT: Magnetometer bias. Use feature pattern: large mean deviation
    with normal variance and normal autocorrelation.

PATTERN C: STUCK VALUE
  ?: some constant value
  ?²: NEAR ZERO
  range: NEAR ZERO
  ??: ~1.0
  ZCR: ~0
  H: ~0
  ? RESULT: Sensor or ADC frozen. All spread measures collapse to zero.
  Easy to diagnose ? the feature vector collapses into a corner.

PATTERN D: HIGH NOISE / EMI
  ?²: HIGHLY ELEVATED
  ??: HIGH
  ZCR: HIGH
  ??: LOW (< 0.5)
  H: HIGH
  ? RESULT: External noise source. The signal is jumping around rapidly.
  Spread measures are all high; smoothness measures are all low.

PATTERN E: SATELLITE TUMBLING
  ?²: VERY HIGH
  range: VERY LARGE (sweeping through full ±60 ?T range for magnetometers)
  ZCR: HIGH
  H: HIGH
  ??: VERY HIGH
  PSD peak: at tumble rate frequency
  Photodiodes also show: HIGH variance, HIGH ZCR
  ? RESULT: Loss of attitude control. Requires immediate safe mode.

PATTERN F: ECLIPSE TRANSITION (photodiodes)
  Mean: DECREASING over segment (eclipse entry) or INCREASING (exit)
  Slope: LARGE (steep transition)
  Range: LARGE (goes from sunlit to dark)
  ??: MODERATE (smooth transition but changing rapidly)
  ? RESULT: Normal eclipse/entry. Must be confirmed with orbital prediction.

4. HOW THE FCNN USES THESE FEATURES
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The FCNN model processes the 18-dimensional feature vector for each
telemetry segment and outputs a probability score ? ? [0,1], where:
  - ? < 0.77: Segment classified as NOMINAL
  - ? ? 0.77: Segment classified as ANOMALOUS

The threshold of 0.77 was selected to maximise F1-score on the validation
set (1,275 training, 319 validation, 529 test samples from OPS-SAT ADCS
data). At this threshold:
  - Test accuracy: 97.54%
  - Precision: 96.30% (96.3% of raised alarms are genuine anomalies)
  - Recall: 92.04% (92% of true anomalies are detected)
  - F1-score: 94.12%
  - ROC-AUC: 99.11%

The high precision (96.30%) means operators can trust that when the model
raises an alarm, it is almost certainly a real anomaly. This is the most
important property for reducing operator alarm fatigue.

The high ROC-AUC (99.11%) means the model has learned a near-perfect
separation between nominal and anomalous feature vectors, regardless of
the chosen threshold.

5. EXPLANATION TEMPLATE FOR OPERATORS
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When the FCNN flags a segment as anomalous, the diagnosis explanation
should follow this template:

"Segment [SEGMENT_ID] from channel [CHANNEL_NAME] at [TIMESTAMP] has been
classified as ANOMALOUS with confidence [SCORE×100]%.

The most prominent anomaly indicators in this segment are:
  - [Feature 1 name]: [value] ? [interpretation, e.g., 'variance is 8×
    above normal, indicating significant signal noise']
  - [Feature 2 name]: [value] ? [interpretation]

Based on the pattern of affected channels and the feature signature, the
most likely root cause is [ROOT CAUSE] with [confidence]% confidence.

[ROOT CAUSE EXPLANATION ? 2-3 sentences explaining the physics]

Recommended corrective actions:
  1. [First action ? most conservative]
  2. [Second action ? if first is insufficient]
  3. [Escalation action ? if anomaly persists]

If this anomaly is not addressed within [timeframe], the consequence may
be [consequence description]."

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END OF FEATURE ENGINEERING AND SIGNAL INTERPRETATION GUIDE
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